

Experiment 33:-

Construct a C program to simulate the optimal paging technique of memory management

Code:

```
#include <stdio.h>

#define MAX_FRAMES 3

void printFrames(int frames[], int n)
{
    for (int i = 0; i < n; i++)
    {
        if (frames[i] == -1)
        {
            printf(" - ");
        }
        else
        {
            printf(" %d ", frames[i]);
        }
    }
    printf("\n");
}

int main()
{
    int frames[MAX_FRAMES];

    // Initialize frames
```

```
for (int i = 0; i < MAX_FRAMES; i++)
{
    frames[i] = -1;
}

int pageFaults = 0;

int referenceString[] =
{
    7, 0, 1, 2, 0, 3, 0, 4, 2, 3, 0, 3, 2
};

int n = sizeof(referenceString) / sizeof(referenceString[0]);

printf("Reference String: ");

for (int i = 0; i < n; i++)
{
    printf("%d ", referenceString[i]);
}

printf("\n\n");

printf("Page Replacement Order:\n");

for (int i = 0; i < n; i++)
{
    int page = referenceString[i];

    int pageFound = 0;

    for (int j = 0; j < MAX_FRAMES; j++)
    {
        if (frames[j] == page)
        {
```

```

        pageFound = 1;
        break;
    }
}
if (!pageFound)
{
    printf("Page %d -> ", page);

    int optimalPage = -1;
    int farthestDistance = -1;
    for (int j = 0; j < MAX_FRAMES; j++)
    {
        if (frames[j] == -1)
        {
            optimalPage = j;
            break;
        }
    }
    if (optimalPage == -1)
    {
        for (int j = 0; j < MAX_FRAMES; j++)
        {
            int futureDistance = 0;
            int foundInFuture = 0;

            for (int k = i + 1; k < n; k++)

```

```
{
    if (referenceString[k] == frames[j])
    {
        foundInFuture = 1;
        break;
    }

    futureDistance++;
}
if (!foundInFuture)
{
    optimalPage = j;
    break;
}
if (futureDistance > farthestDistance)
{
    farthestDistance = futureDistance;
    optimalPage = j;
}
}
}

frames[optimalPage] = page;

printFrames(frames, MAX_FRAMES);

pageFaults++;
```

```
    }  
}  
  
printf("\nTotal Page Faults: %d\n", pageFaults);  
  
return 0;  
}
```

Output:

```
Reference String: 7 0 1 2 0 3 0 4 2 3 0 3 2  
  
Page Replacement Order:  
Page 7 -> 7 - -  
Page 0 -> 7 0 -  
Page 1 -> 7 0 1  
Page 2 -> 2 0 1  
Page 3 -> 2 0 3  
Page 4 -> 2 4 3  
Page 0 -> 2 0 3  
  
Total Page Faults: 7  
  
...Program finished with exit code 0  
Press ENTER to exit console.
```